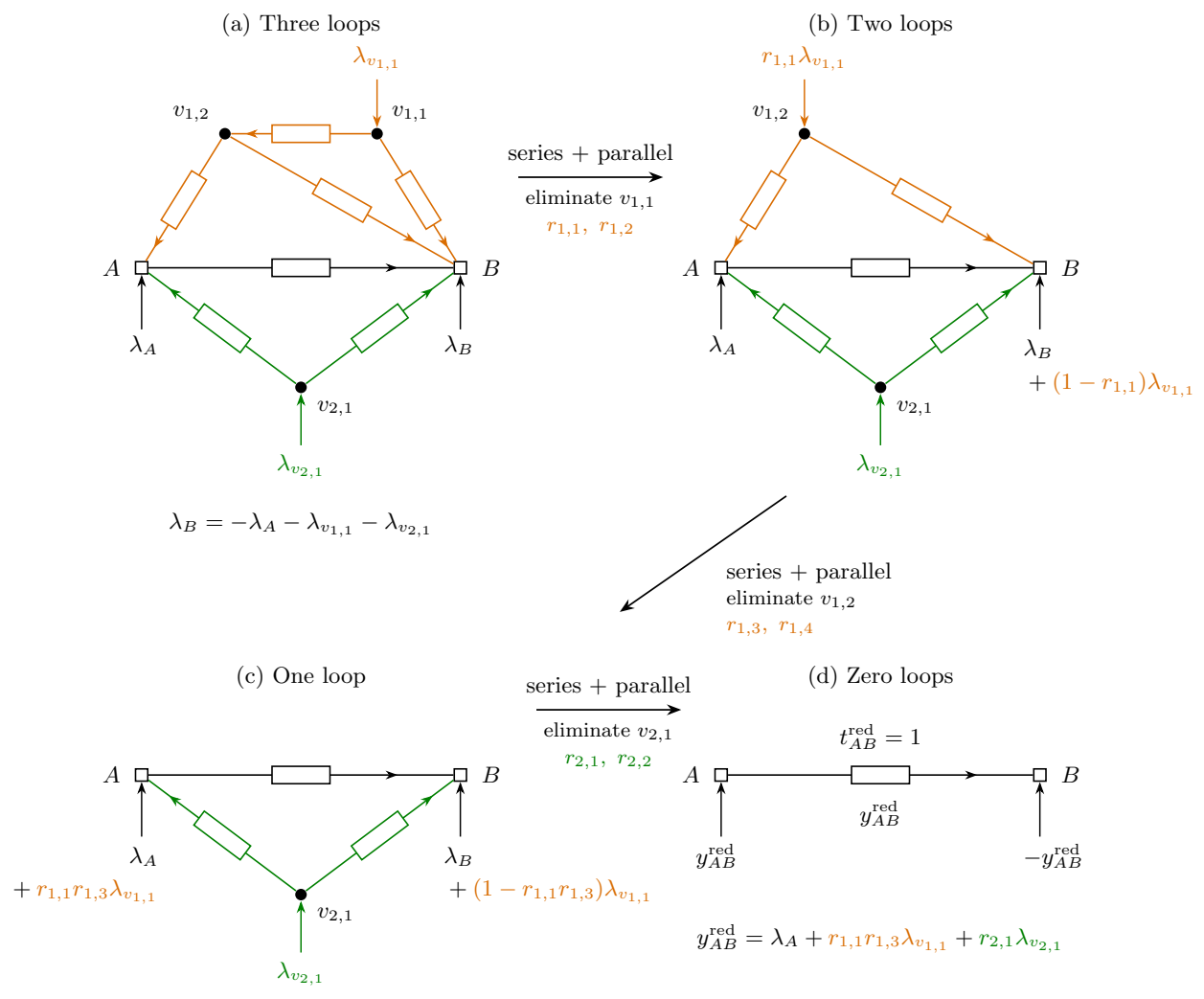

(a) Three loops
$\lambda_{v_{1,1}}$
$v_{1,2}$
$v_{1,1}$
A
B
λ_A
λ_B
$v_{2,1}$
$\lambda_{v_{2,1}}$
$\lambda_B = -\lambda_A - \lambda_{v_{1,1}} - \lambda_{v_{2,1}}$
series + parallel
eliminate $v_{1,1}$
$r_{1,1},\ r_{1,2}$
(b) Two loops
$r_{1,1}\lambda_{v_{1,1}}$
$v_{1,2}$
A
B
λ_A
λ_B
$+ (1 - r_{1,1})\lambda_{v_{1,1}}$
$v_{2,1}$
$\lambda_{v_{2,1}}$
series + parallel
eliminate $v_{1,2}$
$r_{1,3},\ r_{1,4}$
(c) One loop
A
B
λ_A
λ_B
$+ r_{1,1}r_{1,3}\lambda_{v_{1,1}}$
$+ (1 - r_{1,1}r_{1,3})\lambda_{v_{1,1}}$
$v_{2,1}$
$\lambda_{v_{2,1}}$
series + parallel
eliminate $v_{2,1}$
$r_{2,1},\ r_{2,2}$
(d) Zero loops
$t_{AB}^{\text{red}} = 1$
A
B
y_{AB}^{red}
y_{AB}^{red}
$-y_{AB}^{\text{red}}$
$y_{AB}^{\text{red}} = \lambda_A + r_{1,1}r_{1,3}\lambda_{v_{1,1}} + r_{2,1}\lambda_{v_{2,1}}$